

DevOps Assessment - Infrastructure as Code

based on ThoughtWorks' Practical Assessment

July 22, 2022

1 Objective

This practical assessment is intended to:

- show us you can quickly pick up and deploy software stacks.
- show off your DevOps skills and understanding of the requirements.
- show that you can make pragmatic technical tradeoffs given a deadline.
- take around 3-4 hours, please stop if you feel like you are taking longer than 5 hours. We will set up another session to go through what you have done.

2 Background

We have been engaged by a client to build a new website to revolutionise online newsfeeds. Working with us, the client's developers have produced an MVP (Minimal Viable Product) based on a microservices architecture.

At present the MVP is deployed to the client's on-premise test infrastructure, which has now gone down. Your responsibility is to provision a basic execution environment in the cloud to run the MVP.

The client's CIO has given a clear direction that she wants to use Infrastructure as a Service (e.g.AWS, Azure, DigitalOcean, GoogleCloud) rather than Platform as a Service (e.g.Heroku, ECS, AppEngine) as she does not want to get locked into a particular vendor. She perceives that IaaS offers better portability.

Please note: the above is just the outline of the background to give you some context that you should consider in your solution. The actual task for this assignment is detailed in the next section.

3 The Problem

The developers of the system have made the source code available to you on this URL:

<https://github.com/melio-consulting/infra-problem>

The system is written in *Clojure* and consists of three microservices: a UI and two backend services.

- **quotes** which serves a random quote
- **newsfeed** which aggregates several RSS feeds together
- **front-end** which calls the two previous services and displays the results

Note: the **master** branch currently does not work, please use the **develop** branch for this task.

The team's BA has given you the following user story card. It is important to note that you only have maximum 4 hours to work on the user story card. It is your discretion to push back and complete what you can, as long as you can justify (e.g. Maybe the user story scope is not specific enough, what should it be in this case?).

User Story: Cloud deployment environment

As a developer on the project,
I need some infrastructure code to provision an environment in the cloud to test features,
And deploy the microservices,
So that I can verify that the microservices are working

Acceptance Criteria

- **Given** that I have deployed the microservices to a cloud provider
 When I load the application in a web browser
 Then the front end application should be displayed
- Should include good documentation so we can run script from one of our development laptops

You can target a cloud provider of your choice (AWS, Azure or GCP). The cloud environment can be based on Windows or Linux. You should assume developers are able to make use of any Linux, Windows or OSX tooling.

Hint: You can use any Infrastructure as Code tool (such as Terraform, Ansible, Pulumi etc). Remember to provide a good README with good instructions to set it up. The developer may not be well versed in the technology that you have chosen.

4 Outputs

When you are done, you should submit the following items to Harry <harry@melio.ai>:

- A **public git repository** (e.g. GitHub or GitLab) or a **zipped repository** (via email) containing instructions which is capable of setting up the required components with little manual intervention. Please make sure that your submission is void of any credentials and secrets.
- A README or similar file which gives guidance to developers on any necessary dependencies, tools and setup.
- Make some brief recommendations about your approach to 'future work'. I.e. how would you extend what you have done to a productionised system. This could take the form of a document or diagrams. Mention how you would engage with the team.